Capstone 1 - HR Employee Attrition Analysis

Cerlin Immaculate A

Tools: Python | MySQL | Power BI

GitHub: github.com/CERLINIMMACULATEA/Capstone1_HR_Attrition_Cerlin

Business Problem

- **Problem:** Attrition Rate 16.12% (237 of 1470 left).
- **Impact:** High hiring cost, productivity loss.
- **User:** HR Manager & Dept Heads need to know WHY.
- **Objective:** Find Top 3 Root Causes + Give 3 Actionable Solutions.

Dataset

- **Source:** IBM HR Analytics - Kaggle (Link on slide).
- **Size:** 1470 Rows x 35 Columns.
- **Row Meaning:** 1 Row = 1 Employee.
- **Key Fields:** Age, Department, Monthly Income, OverTime, Job Satisfaction (1-4), Attrition (Yes/No).

Project Workflow - Pattern 1

- **Pattern 1:** Python -> MySQL -> Power BI.
- **Python (Pandas):** Cleaning + EDA.
- **MySQL:** 10 Business Questions (GROUP BY, CASE WHEN).
- **Power BI:** Star Schema + DAX + Interactive Dashboard.
- Why 3 Tools? Cleaning in Pandas, Fast Questions in SQL, Story in Dashboard.

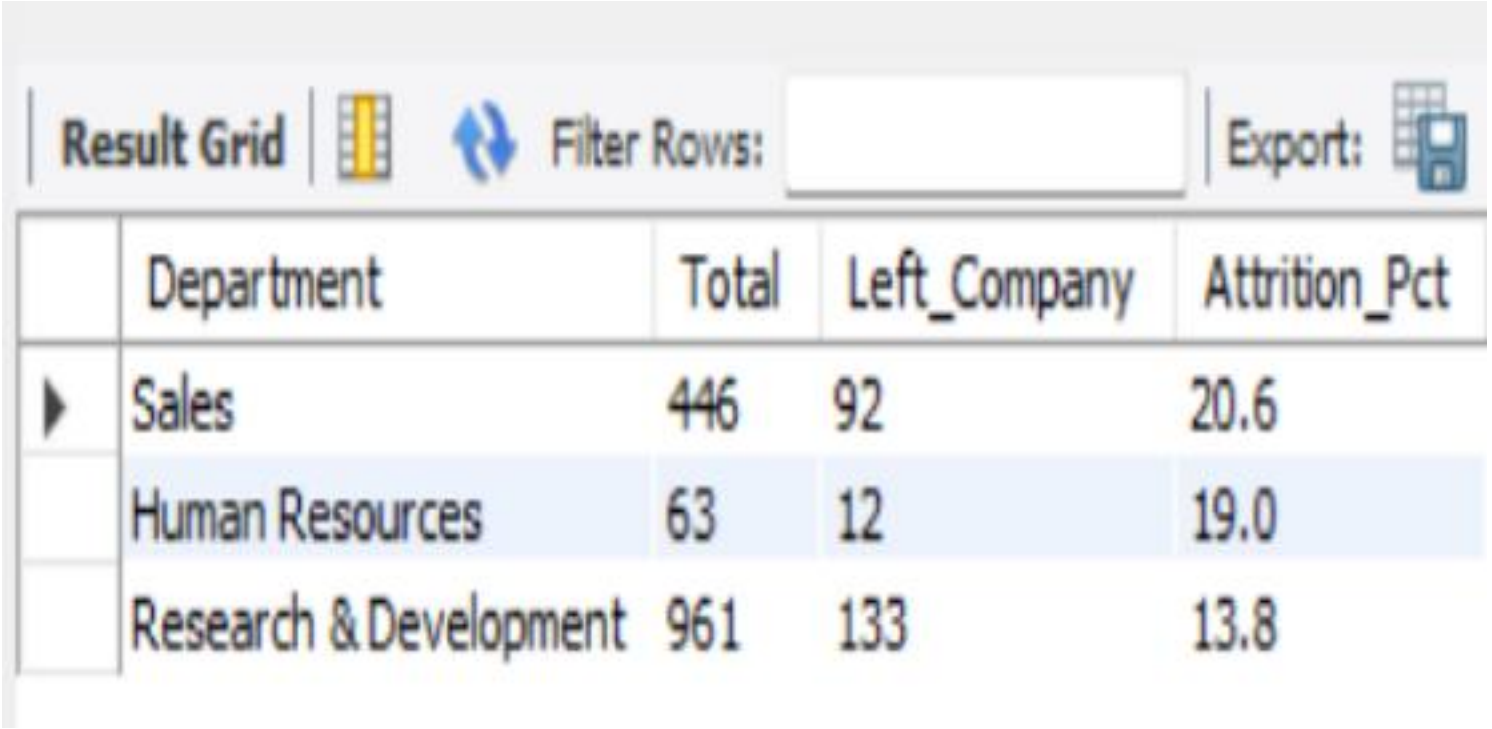
Data Cleaning

- **Before:** Raw CSV with text values
- **Checks:** Duplicates = 0 | Missing Values = 0
- **Fixed:** OverTime Yes/No -> 1/0 using map()
- **Created:** Age Groups, Income Slabs
- **After:** Cleaned CSV - 1470 rows ready (HR_Attrition_Cleaned.csv)
- **Challenge:** Text to number conversion - Solved with Pandas map()

SQL Analysis

- **Q1 Dept wise Attrition:** Sales 20.6% (92/446) - HIGHEST
- **Q2 OverTime:** Yes 30.5% (127/416) vs No 10.4% (110/1054) - 3X Difference
- **Q3 Hike + Satisfaction:** Low Satisfaction (1) + 11% Hike = 22.5% (9/40) vs High (4) = 11.3% (52/459)
- **Full 10 Queries in GitHub:** HR_Attrition_Analysis_Cerlin.sql

Q1 Output : Dept Wise - Sales 20.6% HIGHEST



Department	Total	Left_Company	Attrition_Pct
Sales	446	92	20.6
Human Resources	63	12	19.0
Research & Development	961	133	13.8

Q2 Output : OverTime - Yes 30.5% (127/416) HIGHEST vs No 10.4% (110/1054)

Result Grid | Filter Rows:

	OverTime	Total	Left_Company	Attrition_Pct
▶	Yes	416	127	30.5
	No	1054	110	10.4

Q3 Output : Job Satisfaction - Low (1) 22.8% (66/289) HIGHEST vs High (4) 11.3% (52/459)

Result Grid | Filter Rows: | Export:

	JobSatisfaction	Total	Left_Company	Attrition_Pct
▶	1	289	66	22.8
	3	442	73	16.5
	2	280	46	16.4
	4	459	52	11.3

Data Model - Star Schema

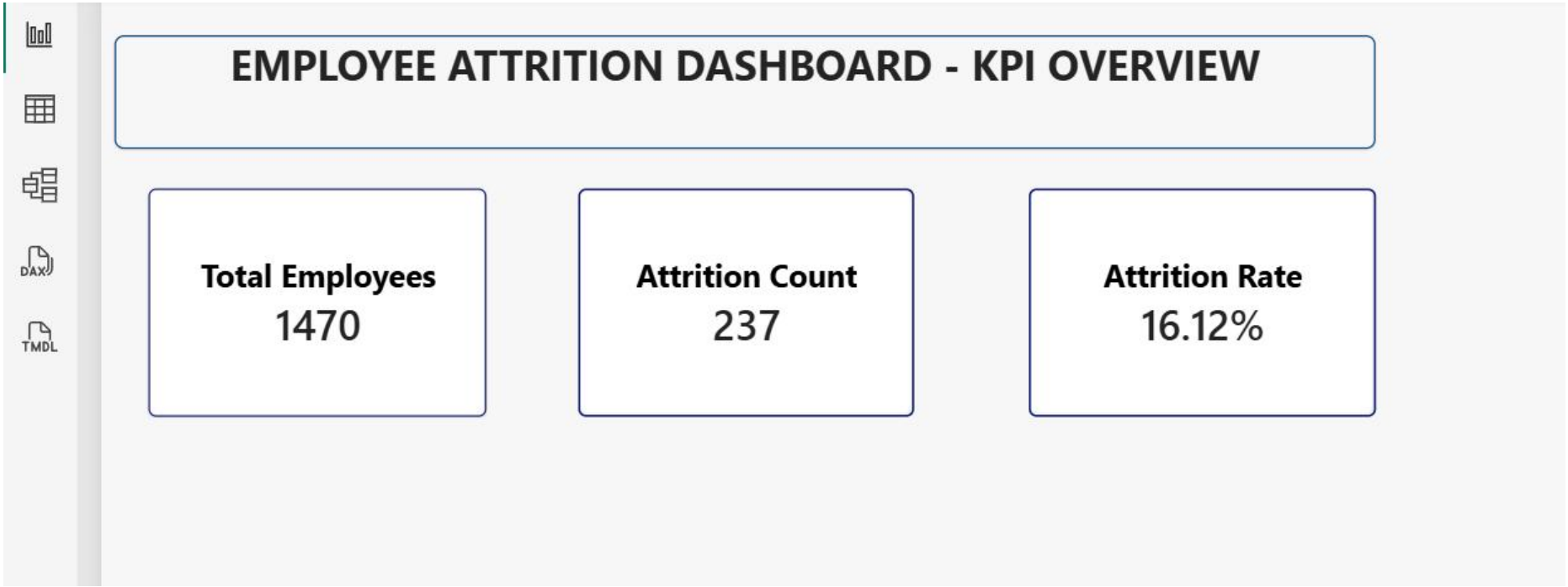
Fact Table
- Fact_Employee - 1470 Rows
- Core Metrics: Attrition, Age, Monthly Income

Dimension Tables
- Dim_Dept → Department, Job Role
- Dim_Education → Education, Education Field
- Dim_Personal → Gender, Marital Status, AgeBand

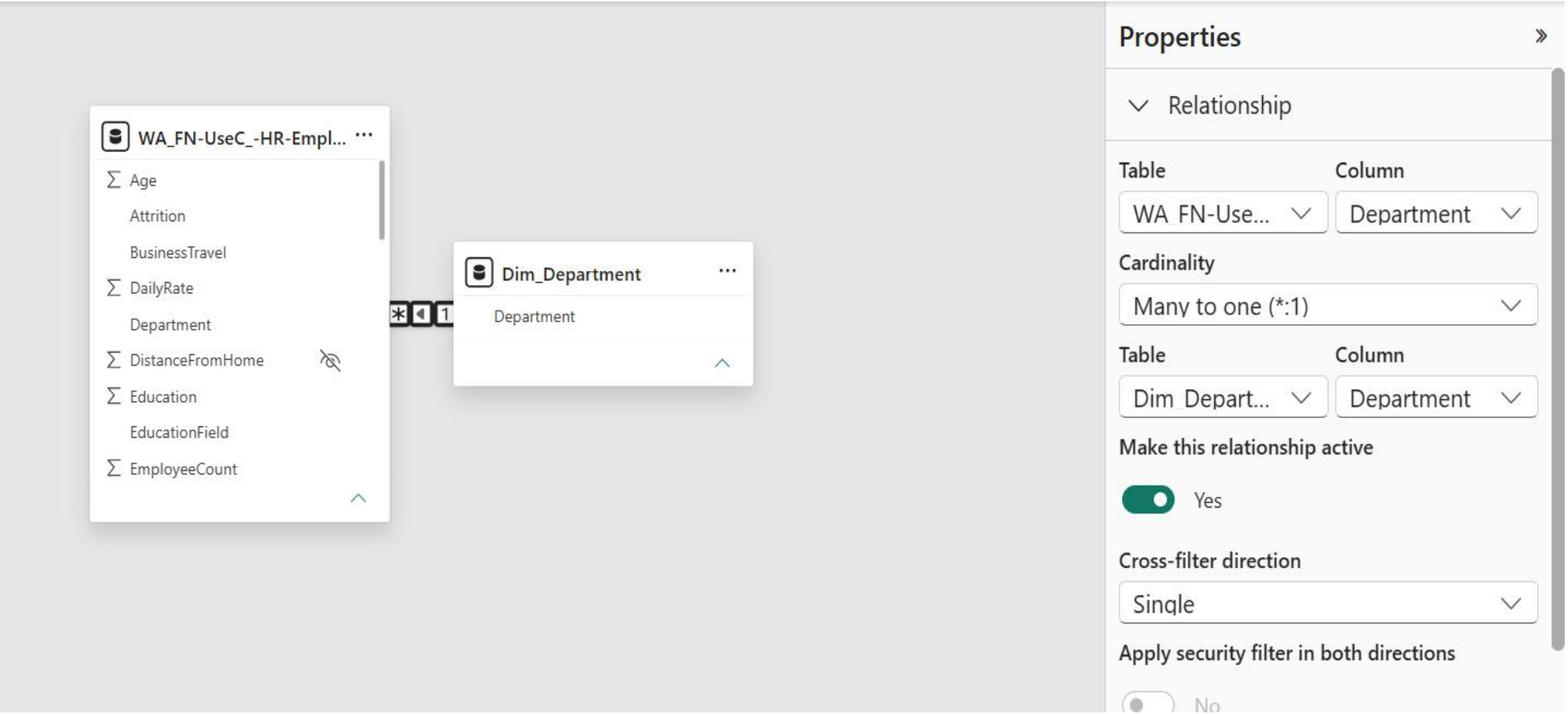
Relationships
- Dim (1) - Fact (Many)
- Dim_Dept - Fact_Employee

DAX Measures
- Total Employee Count = COUNTROWS(Fact_Employee)
- Attrition Count = CALCULATE(COUNTROWS(...), [Attrition]=”Yes”)
- Attrition % = DIVIDE([Attrition Count], [Total Emp Count])

Dashboard Implementation - KPI Overview



Data Model Implementation - Relationship View



Key Insights

Point 1: Overtime = High Attrition (You proved in Q2 - 30.5%).

Point 2: Low Satisfaction + Low Hike = 22.5% Attrition (You proved in Q3).

Point 3: Age 25-34 = Highest Attrition (18.5%) & Sales Dept = 20.8% Attrition.

Recommendations to HR

1. Control Overtime: High overtime = High attrition. Introduce flexible hours & comp-off.

2. Retain New & Low-Income Staff: Focus on 0-2 years tenure group with salary review & clear growth path.

3. Support Sales Team: Revise targets and improve incentive structure for Sales department.

4. Boost Engagement: Launch mentorship & engagement programs for younger, single employees.

Conclusion

- The organization has an overall attrition rate of **16.12% (237 of 1470 employees)**, indicating a critical retention issue.
- Key attrition drivers identified are **excessive overtime, low monthly income, early tenure (0-2 years), and high pressure in the Sales department.**
- **Single and younger employees (18-30 years)** are found to be the most vulnerable group.
- Implementing the proposed targeted HR interventions will help improve retention, reduce hiring costs, and enhance overall workforce stability.



Thank You